

=== UNIT 1: INTRODUCTION TO COMPUTER NETWORKS & PHYSICAL LAYER ===

[Q.1] What is the difference between OSI and TCP/IP reference models?

Difficulty: basic

Category: theory

Tags: osi, tcp-ip, networking, architecture

Punchline:

OSI has 7 theoretical reference layers; TCP/IP has 4 practical implementation layers used in real-wo

Detailed Answer:

1. Architecture:

- OSI Model (Open Systems Interconnection) is an ISO standard developed in 1984 with 7 distinct l
- TCP/IP Model (Transmission Control Protocol / Internet Protocol) was designed by DoD with 4 pra

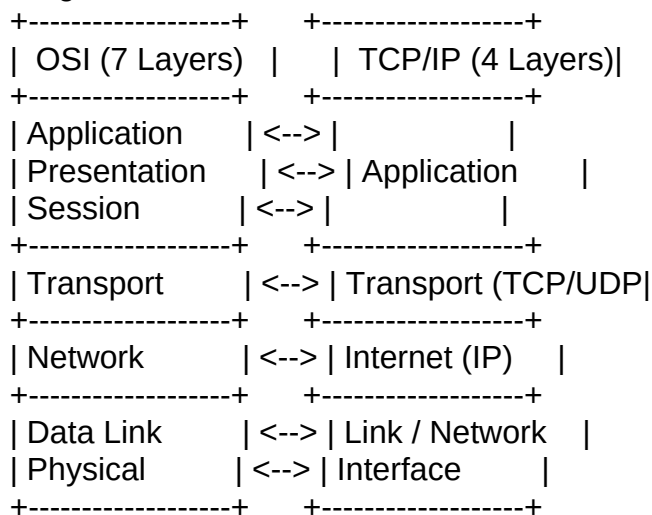
2. Conceptual vs Real-world:

- OSI model strictly separates services, interfaces, and protocols.
- TCP/IP protocols were created first, and the model was just a description of them.

Quick Revision:

OSI is a 7-layer reference model; TCP/IP is a 4-layer working protocol suite.

Diagram:



Follow-up Questions:

Q: Which layer in OSI performs encryption and data compression?

A: The Presentation layer (Layer 6).

Q: At which layer do IP addresses operate?

A: The Network layer (Layer 3 in OSI, Internet layer in TCP/IP).

[Q.2] What is the difference between TCP and UDP?

Difficulty: intermediate

Category: theory

Tags: tcp, udp, transport-layer

Punchline:

TCP is connection-oriented, reliable, and uses a 3-way handshake; UDP is connectionless, lightweight

Detailed Answer:

1. TCP (Transmission Control Protocol):

- Establishes a reliable full-duplex byte stream connection via a 3-way handshake (SYN, SYN-ACK, ACK)
- Features checksums, sequence numbers, flow control (sliding window), and retransmission of lost packets
- Typical use cases: HTTP/HTTPS (Web), SMTP (Email), SSH, FTP.

2. UDP (User Datagram Protocol):

- Unreliable, lightweight, header size only 8 bytes vs 20-60 bytes in TCP.
- Packets are fired independently without acknowledgments or flow control.
- Typical use cases: DNS queries, VoIP (audio calls), Video Streaming, Gaming.

Quick Revision:

TCP = reliable with overhead; UDP = fast and lightweight without guarantees.

Follow-up Questions:

Q: What is the TCP 3-way handshake sequence?

A: 1. SYN from Client -> 2. SYN-ACK from Server -> 3. ACK from Client.

=== UNIT 2: DATA LINK LAYER & MEDIA ACCESS CONTROL ===

[Q.3] What is CSMA/CD and how does it handle packet collisions?

Difficulty: intermediate

Category: theory

Tags: csma-cd, ethernet, collisions, mac

Punchline:

Carrier Sense Multiple Access with Collision Detection listens to the wire before and during transmission

Detailed Answer:

CSMA/CD is the classical MAC protocol used in half-duplex Ethernet:

1. Carrier Sense: Station listens to the wire. If busy, it waits. If idle, it transmits.
2. Collision Detection: While transmitting, the station keeps monitoring the voltage/signal level.
3. Jam Signal: If a collision is sensed, it transmits a 32-bit jam signal so all stations detect the collision.
4. Binary Exponential Backoff: Each colliding station waits a random time interval (0 to $2^k - 1$ slots)

Quick Revision:

Listen before talk, listen while talk; if collision, jam and back off exponentially.

[Q.4] What is the difference between a Hub, a Switch, and a Router?

Difficulty: basic

Category: theory

Tags: hub, switch, router, hardware

Punchline:

A Hub operates at Layer 1 (broadcasts to all ports); a Switch operates at Layer 2 (forwards by MAC address); a Router operates at Layer 3 (forwards by IP address)

Detailed Answer:

1. Hub (Layer 1 - Physical):

- Passive or active repeater. Any frame received on one port is forwarded to all other ports (sin

2. Switch (Layer 2 - Data Link):

- Intelligent multi-port bridge. Maintains a MAC Address Table (CAM) and sends frames directly to

3. Router (Layer 3 - Network):

- Connects different network subnets. Reads IP packet headers and uses routing algorithms (OSPF,

Quick Revision:

Hub = Layer 1 broadcast; Switch = Layer 2 MAC forwarding; Router = Layer 3 IP routing.